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4D printing of shape memory polymers: A comparative study of programming methodologies on various material properties

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Abstract

The emergence of 4D printing has enabled the fabrication of various components that can change in response to external stimuli. Fused filament fabrication is one of the methods for creating shape-changing components using shape-memory polymer (SMP) filament. In order to exhibit the phenomenon of the shape memory effect, programming plays a crucial role. This article discusses two programming concepts, programming during printing (PDP) and programming after printing (PAP), for SMP processed by fused deposition modeling (FDM). We investigated the shape memory properties and other material behavior of PAP and PDP samples considering different thicknesses. We observe that PDP outperforms PAP in terms of shape memory properties based on various characterization tools like Fourier transform infrared spectroscopy (FTIR), x-ray diffraction (XRD), and field emission-scanning electron microscopy (FE-SEM), which are used for macro and microstructural features. Whereas, PAP shows better mechanical properties based on Nanoindentation analysis. The PDP samples achieved a maximum shape recovery of 99.25%, which is 44% higher than PAP for a 4 mm thick sample, and showed a 28% improvement in recovery compared to PAP for a 2 mm thick sample. Statistical analysis reveals significant differences in the means of recovery ratio and shape memory index between PAP and PDP, and no statistically significant difference is found for the fixation ratio. A shape recovery cycle life measurement has been carried out for a PDP bending actuator, which showed recovery until 140 cycles before complete failure. Finally, a working prototype demonstrating effectiveness of PDP and PAP for programming the same SMP in two different ways has been presented.

Keywords: 4D printing, material programming, shape memory properties, smart materials, material characterization, fused deposition modeling, four dimensional printing

(Some figures may appear in colour only in the online journal)

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1. Introduction

The success of a new product in today's market is contingent on its design, materials, production, rapid validation, and optimization to stand out from the crowd. Rapid prototyping (RP) may save time and effort in the early stages of testing a new product [1]. RP can be achieved by various means of conventional and non-conventional methods of manufacturing. Among conventional manufacturing processes, material subtraction methods, including machining and drilling, have drawbacks, such as generating material waste, offering limited design options, requiring longer lead times, incurring higher costs, and having restricted material selection [2]. This led to the development of additive manufacturing (AM) processes. The mechanism of AM involves the addition of different layers of materials, one above the other, to create a final part based on the computer-aided design (CAD) [3]. By building up parts in a precise and controlled manner, AM allows for greater design flexibility and can produce complex geometries that may not be possible using other manufacturing methods [4]. The rapid development of AM has been driven by the growing demand for customized products, improved efficiency, and cost reduction in the manufacturing industry. AM technology offers numerous advantages, including reduced energy and material consumption, greater design flexibility, faster production rates, and reduced assembly requirements, all of which contribute to the trend toward servitization and personalization in manufacturing. Additionally, AM enables mass customization at a lower cost and reduces the risk of excess inventory [5]. AM has made remarkable progress in recent times with the advent of intelligent or smart materials. This technological breakthrough has paved the way for the development of structures that imitate nature and have the ability to self-morph, respond to stimuli, reconfigure, and self-repair. The recent technological breakthroughs have ushered in a novel epoch of opportunities in the realm of engineering and manufacturing, allowing for the creation of innovative products that can adapt to changing environments and heal themselves when damaged [6, 7]. Four-dimensional (4D) printing is an advanced and modern technology that utilizes smart materials to create objects having the ability to alter their characteristic properties over time [8, 9]. This advanced approach uses time as the fourth dimension to make the traditional 3D printing dynamic, enabling the creation of self-assembling or reconfiguring objects that have exciting potential for various applications [10]. In 2013, Skylar Tibbits showcased 4D printing technology at the technology, entertainment, and design conference, demonstrating how printed items can change their design over time in a controlled manner. This innovation involves adding energy and motion to a component to allow it to alter its arrangement, creating stiff and self-assembling structures [11]. Early pioneering works by various researchers in this novel field, have utilized multi material jetting (PolyJet 3D printing) based AM to deposit two different kinds of polymer. The polymers exhibit varying shape memory characteristics, enabling for temporal sequencing and controlled shape-changing activation when subjected to heat. Measurements and simulations of self-folding structures, as well as a simplified reduced-order model that describes the physics of self-folding were also presented [12]. Using the same kind of AM process, active composites were developed for various 4D printing applications. Printing layered structures with various families of shape memory polymer (SMP) fibers, which may be successively triggered to bend as the temperature rises, is used to create the composite structures. The volume percentage of the fibers may be adjusted to regulate the bending deformation. Two prototypes, designed to mimic an insect and a smart hook, have been reported to undergo an extremely complex shape change [13].

Given the latest advancements in this emerging field, 4D printing can be broadly categorized into three distinct classifications. The first type is the embedding of smart materials during the AM process of passive material to create actuated joints or origami structures. Passive materials are those materials that do not respond to the stimulus. Various studies have been conducted to infuse shape memory alloys (SMA) into polymers during fused deposition modeling (FDM) 3D printing. It involves printing the base, pausing the printer to embed the SMA, and then resuming the printing to complete the whole process. This creates an SMA-polymer sandwich composite part with high strength [14, 15]. Smart materials, or the active components incorporated into a 3D printed product, are primarily triggered by external stimuli like heat [16]. Researchers have used Joule's heating to actuate the embedded SMA into the polymer structure to create shape retention bending actuators having variable stiffness and applicable for large reversible deformation [17]. Researchers have also recently integrated conductive polylactic acid (PLA) during the process of 4D printing into silicon and strontium ferrite-based magnetorheological elastomers to develop bistable composite actuators that are lightweight and versatile [18].

The second classification of 4D printing involves the utilization of a single, highly responsive smart material, including but not limited to SMA, hydrogels, and electroactive and SMPs. SMP may be temporarily deformed and fixed in a new configuration; however, when stimulated suitably, these will revert to their original form [17, 19]. Researchers have utilized the response surface approach to create a model for controlling the torsional and flexural deformation in FDM 4D-printed PLA. The central composite design was used to investigate FDM parameters, and its optimization has been achieved [20]. Recently, a single, smart material in the form of polyvinyl chloride processed using FDM has been investigated for its shape memory performance considering various types of programming [21]. A pre-processed composite smart material, which can be 4D printed using a single material AM process, also falls under this second category. Researchers have used UV-light-assisted direct-ink-write AM to create shape memory and self-healing structures from semi-interpenetrating network elastomer composites [22]. Composite materials with a light-responsive shape memory effect (SME) were created by incorporating gold nanoparticles into a polyurethane (PU) matrix, and it was 3D printed using direct ink writing to create intraluminal scaffolds.

The third classification uses multi-material AM of more than one smart material or composites to create 4D printed structures. This approach has the advantage of enhanced flexibility in design and materials for creating shape-changing parts. The smart material can be deposited precisely at predefined places within the passive material to create complicated shape-morphing structures [23, 24]. Through this method, a wide range of intelligent composite structures has been fabricated [25–27].

Thermo-responsive SMPs are a popular choice among the materials utilized in 4D printing. It displays the SME because of the synergistic interaction of two critical factors: net points and switching segments. The net points, which are integrated within the polymer matrix, determine the permanent shape and are responsible for ensuring full recovery and there is an absence of viscous flow. Whereas the switching segments are the ones that undergo a change in mobility depending on the temperature, thus causing the shape change [19]. To quantify the extent of the SME in the SMP, shape recovery ratio (R_r), shape fixity ratio (R_f), and shape memory index (SMI) parameters are generally considered [28]. R_r denotes the ratio between the recovered strain to the deformed strain and provides information on the SMP's ability to recover its original shape. R_f refers to the ratio of the fixed strain to the deformed strain and indicates the material's ability to maintain the deformed shape. SMI, on the other hand, measures the overall performance of the SMP in retaining the deformed shape and recovering its original shape [19, 29]. Extrusion AM is one of the widely utilized processes for 4D printing of polymers due to its scalability, affordability, user-friendly operation, and accessibility of a broad assortment of polymer materials [30]. Researchers have utilized FDM to engineer adaptive metamaterials with self-folding and self-coiling properties through the layer-by-layer programming of SMPs. The study includes a finite element formulation to describe material tailoring, deformation, and experimental demonstrations of the potential for mechanical and biomedical applications such as self-conforming substrates and surgical sutures [31].

Researchers have conducted a detailed investigation of the 3D printing by fused filament fabrication (FFF)/FDM procedure and the impact of various printing process variables on SMP based on PU, as well as their composites with carbon nanotubes. The study effectively illustrates the quality of SMPs in printed specimens and offers a feasible approach for creating filaments and printing them. Sample recovery time was drastically cut when the activation temperature considered was greater than T_g. The activation temperature must be lower than T_g by no more than 10 °C for a reasonable recovery [32]. For the acrylonitrile butadiene styrene-styrene ethylene butylene styrene (ABS-SEBS) composite, its shape memory properties were studied as a function of the deformation temperature. ABS-SEBS blend has a remarkable rubber-like behavior at low and room temperatures while showing a promising SME at high temperatures. The 50/50 blend showcased the best shape recovery and the highest SMI among all the blends tested.

Polylactide (also known as PLA) is an interesting shape-memory material because of its high biocompatibility and biodegradability. The carbohydrates extracted from bio sources like maize, potato, and other wastes, especially vegetables, maybe fermented or chemically synthesized into PLA [33]. It is produced commercially through a two-step process. The first step involves the production of lactide, which is a cyclic dimer of lactic acid. Lactide is typically produced by heating lactic acid under reduced pressure in the presence of a catalyst, such as tin octoate. This causes the lactic acid molecules to undergo intramolecular cyclization and form lactide. In the second step, the lactide monomers are subjected to ring-opening polymerization (ROP), which leads to the creation of PLA polymer chains. The ROP process is typically carried out in the presence of a catalyst, such as stannous octoate, and at elevated temperatures. The resulting PLA polymer has a high molecular weight and can be used for various applications, including packaging materials, textiles, and biomedical implants [34]. For 4D printing, PLA has been widely used for its widespread availability and low cost. Annealing in the programming of SMPs refers to the process of heating and holding the material at a specific temperature to improve its shape recovery behavior. Using PLA of varying thicknesses, cantilever beams have been printed, and their deformation after annealing has been recorded. Each sample's angle of recovery over time was recorded. Functional prototypes, such as cube origami, have been made to demonstrate the potential of the 4D printing prototype [35]. Fe_3O_4, along with PLA, has been utilized to fabricate 4D-printed biocompatible and biodegradable composites. The influence of magnetism on the shape memory properties of the 4D printed parts is investigated. The printed systems may quickly and easily revert to their original forms. A magnetic field of around 27 kHz has been used to activate the composite structure printed to mimic the bone tissue [36]. Through the use of 4D printing, shape memory composites were created by combining PLA and polycaprolactone (PCL). With increasing PCL concentration, the transition temperature (T_g) of PLA/PCL composites drops from 67.2 °C to 55.2 °C, compared with pure PLA. By carefully manipulating the deformation setting, the PLA/PCL shape memory composites undergo controlled sequential deformation with a remarkable SME. The shape recovery process exhibits a response time of fewer than 1.21 s, while the shape fixation and recovery rates exceed 92% [37]. Stress absorbers, as a very complex geometrical component, were printed using PLA in FDM 3D printer. By changing the printing process parameters (like liquefier temperature, layer thickness, and extruder speed) and operational factors (such as activation temperature), which assist in the restoration of the original form, the shape memory qualities of the printed component were assessed. It was found that activation temperature is the most significant parameter responsible for faster recovery [38]. A polymer/paper bilayer was synthesized using PLA and commercial printing paper. Three different composites were made using FDM by varying the printing angles. Actuations of the composite structures were achieved due to varying thermal stresses in a bilayer structure [39]. The FFF was utilized to 3D print porous scaffolds made of 80% PLA and 15% hydroxyapatite. Extrusion was used to generate the composite filament. The mechanical properties,

structural characteristics, and SME were investigated in detail [40].

Programming or training the SMP in 4D printing is a crucial step in order to dictate the shape change in a predefined way. The material programming framework has been developed to assign and tune material characteristics like elasticity and shape change with different magnitudes and anisotropies throughout a volume. This framework can be utilized to illustrate a range of material behaviors [41]. Due to the mechanics of the process involved in FDM 3D printing, the material programming takes place during printing. The pre-strain information stored during the printing affects the shape recovery during actuation [42]. It has been ascertained that smart, responsive structures capable of exhibiting a triple-SME can be fabricated by utilizing the hot-cold programming of SMPs through the FDM/FFF approach. The thermo-mechanical properties of SMPs are tested empirically at elevated as well as reduced temperatures, while the elastoplastic and hyper-elastic behaviors are scrutinized and elucidated in the domain of large deformation. Experimental results show that dual and triple SMPs with self-bending capabilities are feasible [43].

Over the years, there have been many studies on utilizing SMPs and economic processes like FFF/FDM for various 4D printing applications. However, till now, there has been little to no detailed research on the impact of different programming strategies on SMP material properties. Additionally, the available programming methods in 4D printing are limited. To address this gap, a new programming methodology is proposed that can be applied as a post-process to FDM 4D printed parts. A comprehensive investigation and a scientific framework have also been developed to understand the underlying mechanisms of the two programming strategies and their impact on the SME of the 4D printed samples. To determine the most suitable strategy, experimental comparisons are carried out based on spectroscopic, micromechanical, shape memory properties, shape recovery cycle life, and morphological characterizations. Our results demonstrate that both programming strategies offer unique advantages and can be utilized efficiently in various scenarios depending on the user's needs. To determine the statistical significance, Leven's test for variances and t-test for means between different groups is done. Finally, a working prototype has been demonstrated, showing two different programming schemes side by side.

2. Shape programming methodologies

2.1. Conventional programming approach

An integral part of working with SMP is the need for programming or training. The following systematic process is followed for conventional programming:

- In a traditional programming/training process for SMPs, its initial shape is known as a permanent/original shape after being processed.
- Next, a deformation is provided to get a temporary shape of the SMP, which is the fixed shape.
- For the recovery cycle, when the polymer is heated above the transition temperature, it recovers back to its original position [19, 44].

Apart from conventional programming processes, various other methods of programming based on various temperature zones, like cold programming, warm programming, and hot programming [45], can be used. The temperature zones related to the programming and recovery of the SMP with reference to T_g are cold, hypowarm, hyperwarm, and hot. Cold temperature is much below the T_g, hypowarm refers to the temperature just below the T_g, hyperwarm is just above T_g, and hot temperature is much above the T_g.

2.2. Programming during printing (PDP): SME through FDM printing

In PDP, the thermo-mechanical shape programming is done during the process of FDM 3D printing, as depicted in figure 1. The 3D printing process has a specific step-by-step sequence as described:

- The SMP filament is supplied to the heater block in FDM/FFF extrusion AM, where the SMP is heated considerably over its T_g (hot temperature range), known as T_p, causing the polymer chains to become more mobile and allowing them to flow through the nozzle.
- Once the hot filament bead is dispensed from the nozzle, it is deposited on the bed (fixing/loading). The mobility of the polymer chains decreases, and it starts to form physical cross-links or entanglements, which trap the material in a temporary shape. Furthermore, due to the movement of the nozzle, the filament gets stretched during deposition inducing pre-strain in the direction of deposition.
- When the nozzle moves up while depositing layers on top of each other, the lower layers get cooled (room temperature) and solidify. The polymer chains become trapped in a temporary shape, which is characterized by a pre-strained state due to thermal contraction.
- In the recovery cycle, when the FDM processed (PDP) sample is deformed and heated in the range of hyperwarm temperature, due to which the polymer chains become mobile again and enabling them to rearrange themselves into a new conformational state by releasing the pre-strains.

2.3. Programming after printing (PAP)

In PAP, the thermo-mechanical shape programming is done as post-processing after FDM 3D printing, as depicted in figure 2. The complete process of shape programming and recovery is described in a step-by-step sequence:

- The FDM 3D printed SMP sample is first processed as per the hypowarm temperature range (below T_g) such that the chain mobility of the polymer increases and the deformation followed by fixing can be completed without fracture of the sample.

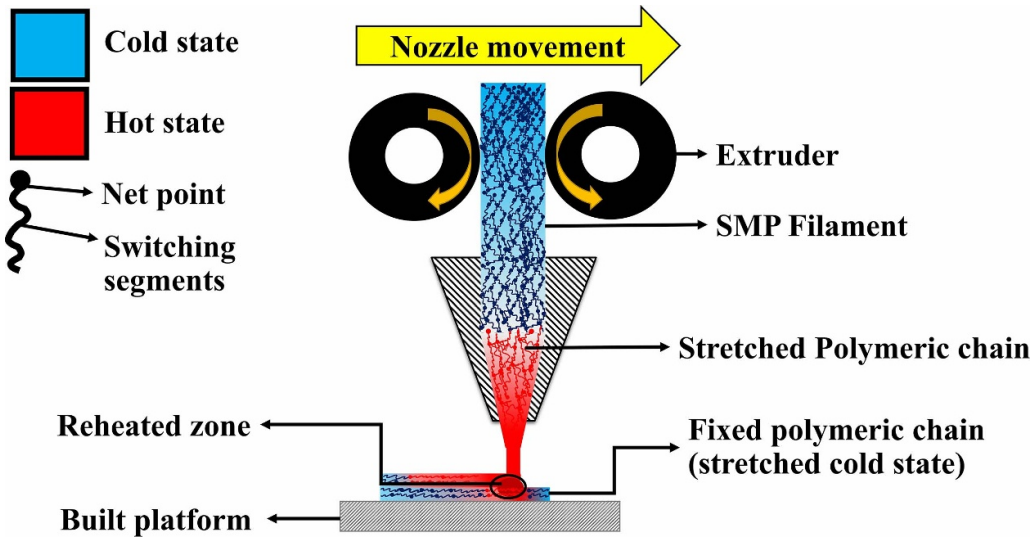


Figure 1. Programming during the printing process (PDP) for FDM/FFF 4D printing.

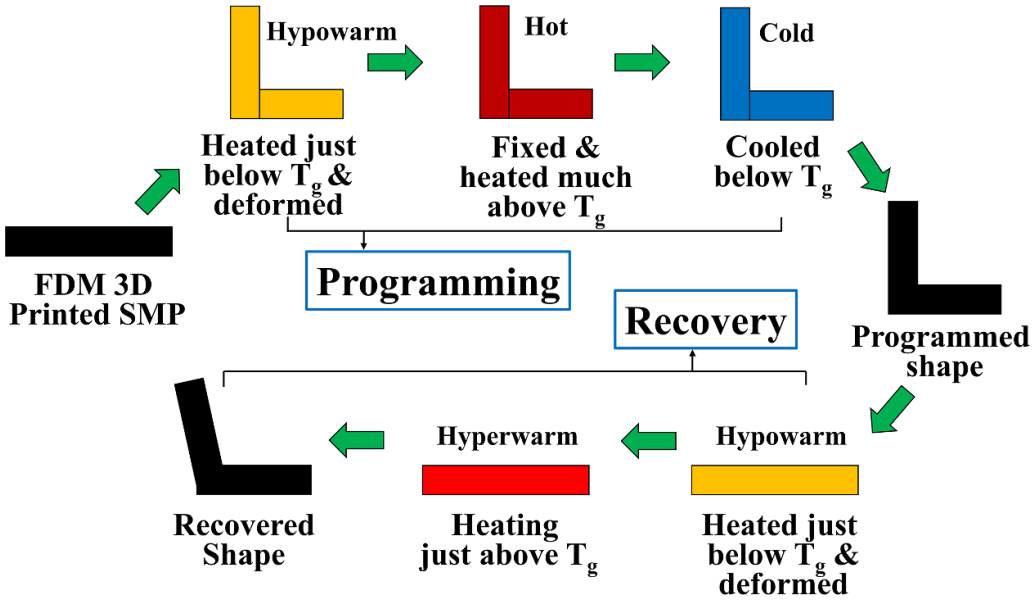


Figure 2. Programming after printing (PAP) process.

- Then the sample, along with the fixture, is heated well above T_g (i.e. hot temperature range), called programming temperature (T_p), where the molecules in the polymer align toward the loading direction due to ordered structures and reduced conformational entropy.
- Keeping the applied stress constant through the fixture, it is then cooled to a temperature less than T_g (room temperature), where the lower conformational entropy is locked due to the decrement in molecular mobility. This completes the programming process, and the sample successfully retains the programmed form as its permanent shape.
- From here, the sample can now be considered for the recovery cycle, where the sample is again heated just below T_g (i.e. hypowarm stage) so that a temporarily deformed shape can be achieved without fracture. Then the sample is heated at T_g or just above T_g (hyperwarm temperature range), for which the shape recovery starts. The recovery starts due to very high molecular mobility, where the molecules tend to achieve higher entropy. Therefore, the molecules in an ordered configuration are changed to the coiled configuration releasing the stored stress and deformation and leading to shape recovery.

Among the above-mentioned programming methodologies, PDP and PAP have been considered for a comparative study based on their various material property change due to the effect of thermo-mechanical programming.

Table 1. Various properties of PLA-SMP.

Properties	Values
Transition temp (°C)	60–62
Hardness (Shore D)	93
Filament size (mm)	1.75
Tension (N)	118–147
Elongation before break (%)	4–5
Density (kg m^{-3})	1250
Moisture absorption (%)	0.6
Melting point (°C)	170–180

3. Materials and method

3.1. Material

For the experiment, PLA, commonly known as polylactide or PLA, has been considered as the SMP. It is made from a biodegradable material, starch extracted from corn, potato, sugar cane, and tapioca. PLA is classified as a semicrystalline thermoplastic that encompasses both ordered and disordered components. During times of stress, the structured structures serve as anchoring sites for shape fixing, and the extended polymeric chains that lack any ordered structure serve as the flexible and elastic component that absorbs strain energy throughout the deformation cycle. Once the deformation is completed, the addition of heat enables the long chains in the polymer to bounce back from their distorted condition, releasing any residual pre-strain and enabling the deformed shape to recover. PLA in the form of a filament of diameter 1.75 mm is procured from WOL 3D Pvt. Ltd [46]. The characteristics of the material furnished by the manufacturer are provided in table 1.

3.2. 4D printing of PLA-SMP

A 3D model of the part is designed in commercial CAD software and saved as an standard tessellation language (STL) file, so it may undergo further preparation. The Simplify3D software is used to slice the model, and a G-Code file is created for 3D printing. The FFF/FDM 3D printer considered for the experiment is Ender 3 (Shenzhen Creality 3D Technology Co. Ltd.). It is manually leveled and well calibrated before the actual printing. The samples are divided into different sets by thickness and programming scheme. In set 1, samples have a thickness of 4 mm, and in set 2, samples are 2 mm thick, as shown in figure 3. The raster angle considered for the present study is +45, −45 refers to the angle at which the nozzle deposits the material layer by layer. Table 2 represents the default process parameters considered for the present study.

3.3. Shape memory property characterization

One of the convenient, economical, yet robust methods for characterizing the shape memory properties (R_r, R_f, and SMI) is utilizing the fold deployment test [37, 47, 48]. R_f refers to the ratio of the fixed strain to the maximum programmed strain

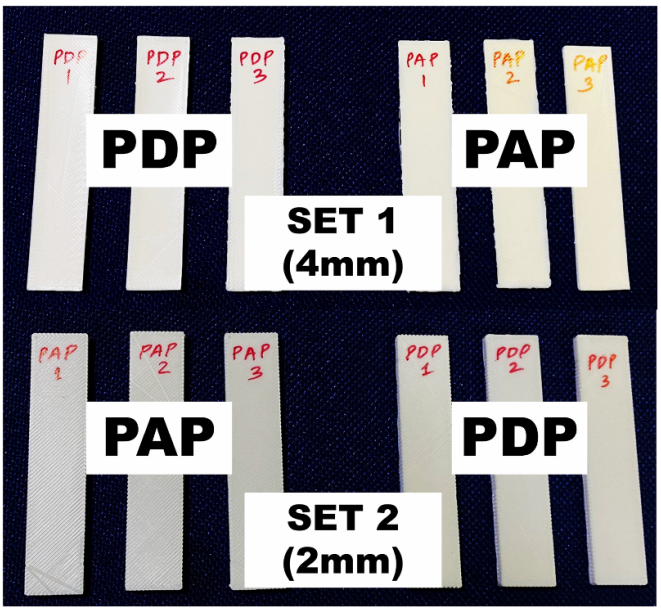


Figure 3. FDM 4D printed samples.

Table 2. Standard settings used for FDM/FFF for both PDP and PAP.

Factors	Values
Extruder movement speed (mm s^{-1})	50
Layer thickness (mm)	0.12
Liquefier temperature (°C)	200
Raster angle (°)	+45, −45
Nozzle diameter (mm)	0.4
Infill percentage (%)	100
Built platform temperature (°C)	60
Speed of retraction (mm s^{-1})	25
First layer thickness (mm)	0.3

and indicates the material's ability to maintain the fixed shape. R_r denotes the ratio between the recovered strain to the fixed strain and provides information on the SMP's ability to recover to its programmed shape. SMI, on the other hand, measures the overall performance of the SMP by considering both R_f and R_r. The process of fold deployment characterization depends upon an accurate measurement of various angles. θ_{max} is the maximum angle obtained after printing the part using an FDM 3D printer. θ_{fix} is the fixed angle that represents the position of the part after being programmed. θ_{def} refers to the deformation angle provided after the programming process as a temporary deformation. θ_{end} is the final angle after the shape recovery. Fold deployment shape memory characterization has been performed for both PDP and PAP samples. The detailed method of characterization is shown in figures 4 and 5. The following equations (1)–(4) are required for the shape memory property calculations,

$$R_f = \frac{\theta_{fix}}{\theta_{max}} \tag{1}$$

where R_f is the fixation ratio, θ_{fix} is the fixed angle, and θ_{max} is the maximum angle.

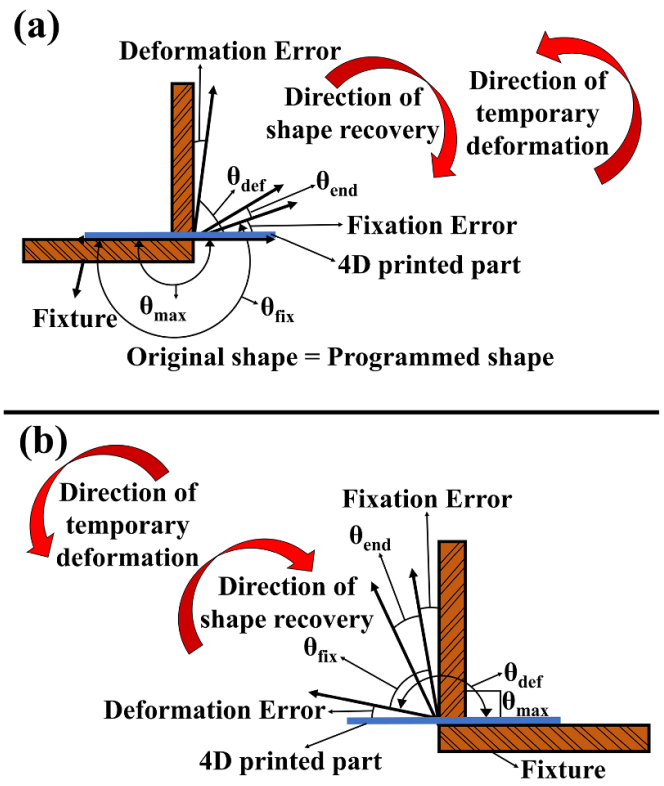


Figure 4. Angle measurement scheme for SME characterization (a) PDP (b) PAP.

Shape recovery ratio for PDP:

$$R_r = \frac{\theta_{fix} - (\theta_{end} + (90 - \theta_{def}))}{\theta_{fix}} \quad (2)$$

where R_r is the recovery ratio, θ_{fix} is the fixed angle, θ_{end} is the recovered angle, and θ_{def} is the deformation angle. All the angles are considered only for the PDP process.

Shape recovery ratio for PAP:

$$R_r = \frac{\theta_{fix} - (\theta_{end} + (180 - \theta_{def}))}{\theta_{fix}} \quad (3)$$

where R_r is the recovery ratio, θ_{fix} is the fixed angle, θ_{end} is the recovered angle, and θ_{def} is the deformation angle. All the angles are considered only for the PAP process,

$$\text{SMI} = R_r \times R_f \quad (4)$$

where SMI is the shape memory index, R_r is the recovery ratio, and R_f is the fixation ratio.

3.3.1. Fixture for fold deployment characterization. A wooden fixture has been designed to provide temporary deformation in the case of PDP and a fixed state for PAP. The designed fixture (figure 5) has a 90° hinge, due to which the sample deforms only at 90°, and it remains the same for all the samples throughout, reducing the scope for error in the measurements. For PDP, the fixture is used to provide only the deformation angle, i.e. θ_{def}, and for PAP, it is used for both the fixed angle, i.e. θ_{fix} (the fixture with the samples goes inside the hot water) and deformation angle. For mounting the sample onto the fixture, nylon self-locking cable zip ties are used.

3.3.2. Hot water actuation testing apparatus. An in-house hot water actuation testing apparatus (figure 5) has been developed for testing the 4D-printed SMP to get various shape memory properties. The advantage of using hot water is that it provides heat to the polymer sample homogeneously. A relay circuit-based temperature controller has been utilized to maintain precise control over the water temperature, with an accuracy of ±1 °C. A temperature probe is positioned nearer to the sample to provide the temperature value accurately. A mechanical stirrer is utilized to distribute heat from the heating element throughout the glass tank. For accurate measurement of recovery angles, a video analysis and modeling software (Tracker) [49] has been utilized where each of the individual frames can be selected, and the change in angle or shape of the 4D printed part during the actuation can be determined.

3.3.3. Angle measurement schemes for PAP and PDP. Various angles must be measured to utilize different equations (1)–(4) for getting the R_r, R_f, and SMI, respectively. Pilot experimentations were carried out to determine the hypowarm temperature of around 50 °C, hyperwarm temperature of 70 °C, and hot temperature of 90 °C. For the case of PDP (figure 4(a)), the thermo-mechanical shape programming is done during the printing process (as explained earlier in section 2.2). The programmed shape is the same as the printed shape, i.e. denoted by θ_{max} (180°). But due to the variation in printing process parameters affecting the dimension, there may be some errors between the printed and fixed shapes. Therefore, considering the fixation error, the final fixed angle can be denoted as θ_{fix}. The sample is temporarily deformed by heating below its T_g (around 50 °C), then mounted on the fixture to bend at 90°, and set for 10 min. After that, the sample is removed, and the angle is measured. There may be variations between the required and actual deformation angles, called deformation errors. The final deformed angle is denoted as θ_{def}. Once deformed, the sample is tested for its shape recovery using the hot water actuation testing apparatus at a temperature well above the T_g (around 70 °C), during which it recovers back, and the angle is measured to be θ_{end}.

For the case of PAP (figure 4(b)), the thermo-mechanical shape programming is done as post-processing after the sample has been printed. The sample is first mounted on the fixture and bent to 90° (θ_{max}) by providing a preheat temperature of 50 °C. The fixture, along with the mounted sample, is then heated to the T_p (around 90 °C) for 10 min. When removed from the heated bath and cooled and the angle is measured. This angle is a fixed angle (θ_{fix}) after considering the error related to it called fixation error. Again the sample is bent, and deformation is provided. This deformation is temporary and is termed as θ_{def} after considering the error associated with it called the deformation error. Once it is completed,

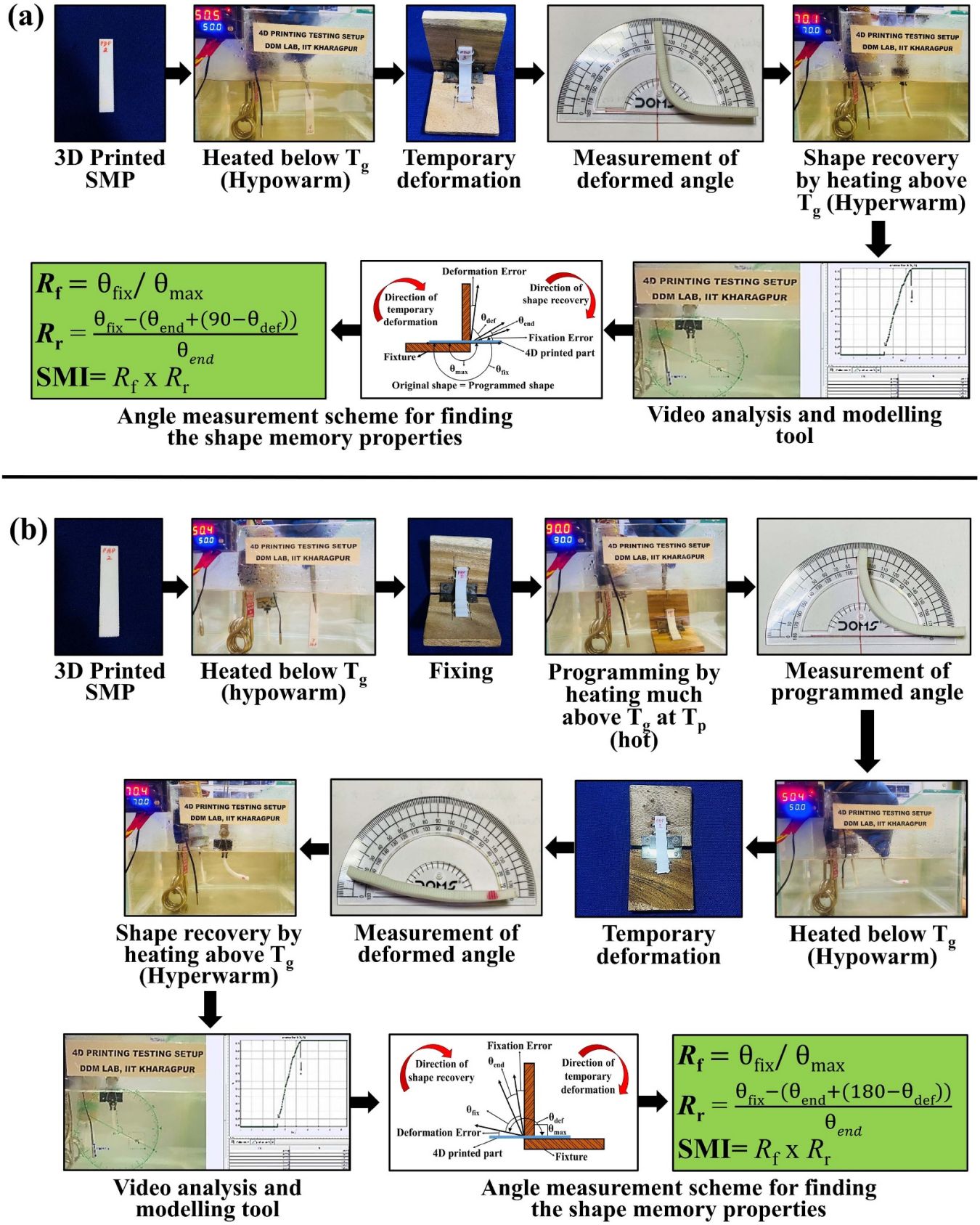


Figure 5. Shape memory characterization method (a) PDP (b) PAP.

now the PAP samples are characterized in the setup at around 70 °C. It recovers and provides the final position. The angle associated with it is θ_{end}.

3.3.4. Statistical analysis. The statistical analysis has been conducted using a Student's t-test, also known as a t-test. The t-test is a crucial statistical method employed to determine whether the difference in means between two groups is statistically significant or not [50]. There exist three variations of t-test, including the one-sample t-test, independent-samples t-test, and paired-samples t-test [51]. For the present study, independent samples t-test has been conducted between the R_r, R_f, and SMI for PAP and PDP samples separately for both the sets, using IBM SPSS statistical software package (Version 26.0, 64-bit edition, IBM Corp, Armonk, NY, US). A Levene's test for equality of variance has been performed to decide the assumption of equal variance and non-equal variance for the t-test [52]. A significance level (α) of 0.05 is considered for both Levene's test and the t-test. A null hypothesis that the variance is not equal and its alternative hypothesis has been considered for Levene's test. Alternatively, the null hypothesis states that there is no significant difference between the means of PAP and PDP, and its corresponding alternative hypothesis is considered for the t-test. The probability, $p < 0.05$, has been considered to accept the null hypothesis, and $p > 0.05$ has been considered to reject the null hypothesis and accept the alternate hypothesis.

3.4. X-ray diffraction

X-ray diffraction (PANalytical EMPYREAN) with Cu-Kα radiation at a rate of 2° min^{-1} has been utilized to study the difference in crystallinity peaks for PAP and PDP samples. The percentage of crystallinity (C (%)) has been determined using the peak integration function in Origin Pro 2022b graphing and analysis software package and then substituting the values in equation (5),

$$\mathrm{C}(\%) = \frac{\text{Area of crystalline peaks}}{\text{Area of all peaks}} \times 100. \tag{5}$$

3.5. Fourier transform infrared (FTIR) spectroscopy

The infrared spectra of the differently programmed samples, i.e. PAP and PDP, were acquired with a Bruker Lumos II FTIR spectrometer in the 4000–550 cm^{-1} interval at a resolution of 4 cm^{-1}, attenuated total reflectance (ATR) mode with ten scans per spectrum. The optical measurement process starts with the contact of Germanium (Ge) ATR crystal tip having a diameter of around 100 μm onto the pre-determined focused spots in the sample. OPUS software is utilized for the evaluation and visualization of the resulting data. The baseline correction is done using the concave rubber band method with 64 points. The smoothing has been carried out using Savitzky–Golay filter.

3.6. Nanoindentation analysis

The plot between load and penetration depth showing the responsiveness of a material to an indentation load is a well-known property of the nanoindentation technique. This essentially reflects the microscale deformation of the material. Therefore, nanoindentation measurements were performed on PAP and PDP samples using a Hysitron TI 950 TriboIndenter (Brukers Corporation, USA) having a diamond Berkovich-type indenter tip with a diameter of approximately 100 nm. Load control mode has been utilized, and the load versus displacement curves were recorded, and the effective hardness and modulus values can be derived from them by using Oliver and Pharr method [53]. Ten different spots on the sample surface were indented at a given load of 8 mN, a rate of 0.35 mN s^{-1}, at a depth of 3 μm. The loading, dwelling, and unloading times have been considered 10 s, 5 s, and 10 s, respectively.

3.7. Shape recovery cycle life

Shape recovery cycle life refers to the number of times a shape memory material can undergo its shape recovery cycle without losing its SME and complete failure [54]. Shape recovery cycle life measurements are carried out by continuing the fold deployment actuation tests discussed in section 3.3 repeatedly. Only PDP samples from set 1 are considered for the cycle life characterization as PDP shows superior shape memory properties than that of PAP, as depicted in section 4.1. The sample is deformed using a fixture and recovered in hot water repeatedly till it fails completely, and in each step, the recovery properties are calculated.

3.8. Fractography study by field emission-scanning electron microscopy (FE-SEM)

The morphology of the cross-section of the fractured surface from the shape recovery cycle life specimen was investigated through SEM (FE-SEM, LYRA3 TESCAN). Nano coating of gold/palladium has been deposited on SMP before SEM to make it conductive.

4. Results and discussion

4.1. Shape memory properties

The shape memory properties have been determined for two sets of PAP and PDP samples, where there is a difference in the thickness between the two sets. It can be observed from figure 6 and table 3 that for both the sets, PDP showed better R_r than that PAP, and the probable reason may be because the release of pre-strain accumulated during the printing process is more. Whereas for PAP samples, the pre-stains induced during printing are released first before the thermal strain accumulation during the PAP process. The maximum R_r achieved is around 99.25%, observed for the 4 mm thick PDP sample, which is approximately 44% more than the corresponding PAP

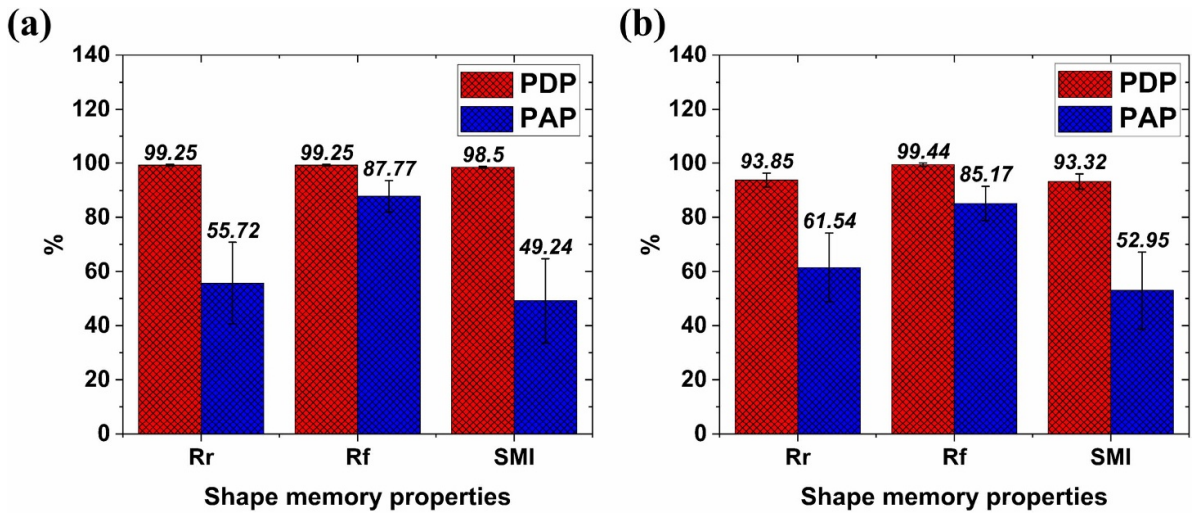


Figure 6. Shape memory property difference between PAP and PDP (a) set 1: 4 mm thickness and (b) set 2: 2 mm thickness.

Table 3. Shape memory properties for PAP and PDP.

Programming during printing (PDP)							
SET 1, samples ($t = 4$ mm)	θ_{max}	θ_{fix}	θ_{end}	θ_{def}	R_f (%)	R_r (%)	SMI (%)
1	180	179	1	90	99.44	99.44	98.88
2	180	179	0	88	99.44	98.88	98.32
3	180	178	0	89	98.88	99.44	98.32
Average					99.25	99.25	98.50
SET 2, samples ($t = 2$ mm)	θ_{max}	θ_{fix}	θ_{end}	θ_{def}	R_f (%)	R_r (%)	SMI (%)
1	180	178	10	90	98.88	94.38	93.32
2	180	179	11	85	99.44	91.06	90.55
3	180	180	7	90	100.00	96.11	96.11
Average					99.44	93.85	93.32
Programming after printing (PAP)							
SET 1, samples ($t = 4$ mm)	θ_{max}	θ_{fix}	θ_{end}	θ_{def}	R_f (%)	R_r (%)	SMI (%)
1	90	77	17	150	85.55	38.96	33.33
2	90	75	10	160	83.33	60.00	50.00
3	90	85	12	165	94.44	68.20	64.40
Average					87.77	55.72	49.24
SET 2, samples ($t = 2$ mm)	θ_{max}	θ_{fix}	θ_{end}	θ_{def}	R_f (%)	R_r (%)	SMI (%)
1	90	70	25	168	77.77	47.14	36.66
2	90	80	10	167	88.88	71.25	63.32
3	90	80	15	168	88.88	66.25	58.88
Average					85.17	61.54	52.95

sample. If we analyze the results for set 2, i.e. sample thickness of 2 mm, R_r for PDP is around 28% more than that of PAP, suggesting that the lower the thickness, low is the pre-strain stored during printing, and PAP is also more effective. Comparing the values of R_f for both sets of samples, PDP shows better fixation as the molten polymer is deposited onto the built plate, thus having better adhesion and hence fixation. Still, on the other hand, the fixation of the PAP sample is done by using a fixture and then heating it above T_g but below the melting temperature. Therefore, PDP shows a better R_f of about 14% more than that PAP. Similarly, SMI, which considers both the R_r and R_f, is more for PDP than that of PAP, and it is almost the same for both sets having different thick samples.

For set 1, Levene's test for R_r and R_f suggests variance homogeneity cannot be assumed for the t-test, whereas for SMI, homogeneity of variance can be anticipated. The t-test

for R_r shows a p-value (0.038) that is less than α, indicating there is a difference between the means of PAP and PDP for R_r, which is statistically significant. However, the t-test for R_f shows a p-value (0.077) that is more than α, which depicts no difference between PAP and PDP means for R_f being not statistically significant. For SMI, the t-test shows a p-value (0.005) that is less than α, indicating a statistically significant difference between the means of PAP and PDP for SMI.

For set 2, Levene's test for R_r and SMI suggests variance homogeneity cannot be assumed for the t-test, whereas for R_f, homogeneity of variance can be anticipated. The t-test for R_r shows a p-value (0.013) that is less than α, indicating the difference between the means of PAP and PDP for R_r, which is statistically significant. However, the t-test for R_f shows a p-value (0.06) that is greater than α, suggesting that the difference between the means of PAP and PDP for R_f is not statistically significant. For SMI, the t-test shows a p-value (0.035) that is less than α, indicating a difference between the means of PAP and PDP for SMI, which is statistically significant.

Overall, it is found that there is a difference between the means of PAP and PDP for R_r and SMI, which is also statistically significant but not for R_f. Therefore, more samples should be tested in the future, and further statistical analysis should be carried out to find differences between PDP and PAP for the fixation ratios where no significant difference was found.

4.2. X-ray diffraction

The x-ray diffraction analysis for both PDP and PAP is shown in figure 7, where it can be clearly distinguished that the PDP shows a halo peak suggesting amorphous nature along with few crystalline peaks (27.43° and 29.39°) for the semicrystalline PLA. PAP shows much more crystalline peaks as compared to PDP. The characteristic peaks corresponding to stable α-form are 16.66° (200)/(110), 18.9° (203), 22.22° (015), and 29.3° (216). The peak corresponding to β-form is obtained at 27.43°. These characteristic peaks of PLA are well reported in various literature [55, 56]. The crystallinity percentage obtained using equation (5) for PDP and PAP are 6.5% and 42%, respectively. The difference in crystallinity between the two programming processes validates the better SME observed in PDP than that of PAP. In PDP, there is a combination of long polymeric chains (high) and net/hard points (low). The chains greatly help elastic strain recovery, thereby showing better shape memory properties. In the case of PAP, the hard points or the crystalline regions are more; therefore, the polymeric chains cannot recover completely, thus showing low shape memory properties. The x-ray diffraction peak difference between PDP and PAP is in good agreement with the nanoindentation results.

4.3. FTIR spectroscopy

The ATR-FTIR spectra obtained for PAP and PDP samples are presented in figure 8. The –C–H stretching modes are obtained at 2993 cm^{-1}–2856 cm^{-1}. The stretching band (–C=O) has been accepted at around 1749–1751 cm^{-1}. The

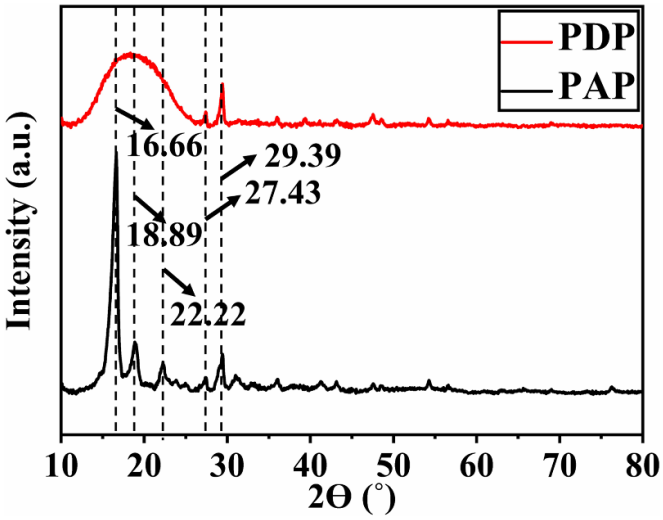


Figure 7. X-ray diffraction peaks for PDP and PAP samples.

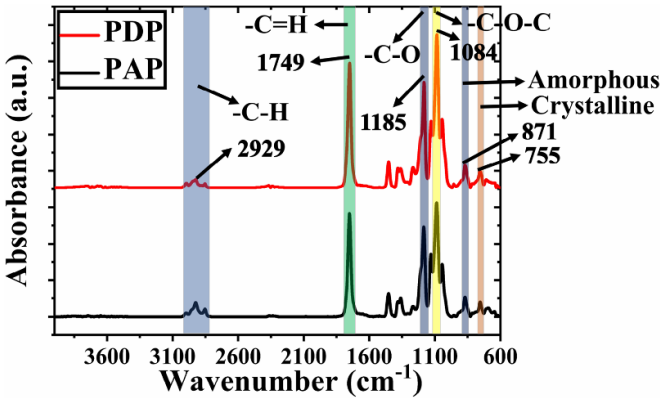


Figure 8. FTIR spectroscopy peaks for PDP and PAP samples plotted in ATR mode.

ester (–C–O–) symmetric stretch is obtained at ∼1185 cm^{-1}, and the (–C–O–C–) asymmetric stretch is at ∼1085 cm^{-1}. The peaks ∼1360 cm^{-1}–1380 cm^{-1} correspond to –C–H bending. Peaks ∼1452 cm^{-1}–455 cm^{-1} correspond to CH_3 asymmetric bending. All the observed peaks are well stated in the literature for the semicrystalline PLA [57]. Apart from these, the two important peaks ∼855 cm^{-1}–871 cm^{-1} and 755 cm^{-1} correspond to the amorphous and crystalline PLA, respectively [58]. For PAP, the amorphous peak shifts from 855 cm^{-1} (PDP) to 871 cm^{-1}. These peaks are ascribed to C–C backbone stretching and rocking mode. The change to the wavenumber is due to the crystal lattice's dipole–dipole interaction, thus suggesting the formation of a crystalline structure [59]. This is because, during PAP's reprogramming, certain disordered components turned into crystals, thereby showing less shape memory properties than PDP. This is in complete agreement with the results so obtained in x-ray diffraction results. Moreover, the crystallinity index, i.e. the ratio between the areas under the crystalline peak ∼755 cm^{-1} and the amorphous peak ∼872 cm^{-1} of PAP, is found to be 78.3%, which is more than that of

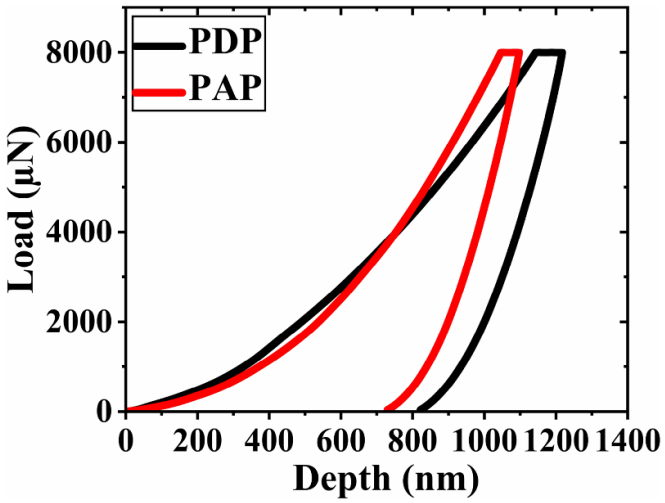


Figure 9. Load vs. depth plot for PDP and PAP.

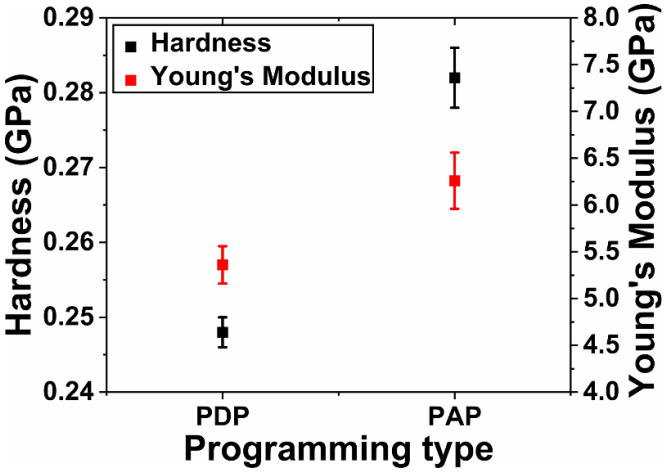


Figure 10. Hardness and Young's modulus plot obtained from nanoindentation for PAP and PDP samples.

PDP. The crystallinity index of PDP obtained by the ratio between the areas under the crystalline peak ∼755 cm^{-1} and the amorphous peak ∼869 cm^{-1} is 59%.

4.4. Nanoindentation analysis

The load vs. penetration depth for PDP and PAP is depicted in figure 9. The depth of penetration during the dwelling of 5 s is about 0.6 nm for PAP, whereas, for PDP, it is around 4.6 nm. This is due to the high crystallinity in PAP, also previously recorded through x-ray diffraction and FTIR analysis. This suggests that the hard region (or net points) increased for the PAP SMP sample is more than that of PDP. Therefore, this also proves the reason for showing low shape memory properties by PAP samples compared to PDP samples. As shown in figure 10, the hardness and Young's modulus values for PAP are around 0.28 GPa and 6.25 GPa, respectively, which is more than that of the PDP, i.e. 0.245 GPa and 5.4 GPa, respectively.

4.5. Shape recovery cycle life

As the best SME is shown for the PDP sample, it should be tested for its viability as a low-cost, low-cycle thermally actuated shape-changing structure for various applications. During the cyclic tests, there is a great chance of structural and functional failure. Therefore, the shape recovery ratio for each cycle has been calculated, and the fractured surface has been presented in figure 11. The PDP sample could actuate till 140 cycles before final failure by providing a recovery of around 60%. Figure 12 shows the graph between the R_r and the actuation cycle. From figure 11, it can be seen that the first crack was generated by the 15th cycle. Then the rupture propagated through the sample's breadth and penetrated layer by layer. It can be observed in a few cycles, the R_r value increased, like in the case of the 90th cycle. It is because the portion of the part under the loading and unloading cycle is made up of a layered structure, and when the first layer got damaged till the 89th cycle, the heat required for the actuation for the thinner part is much more sufficient than before, and hence, there is a slight

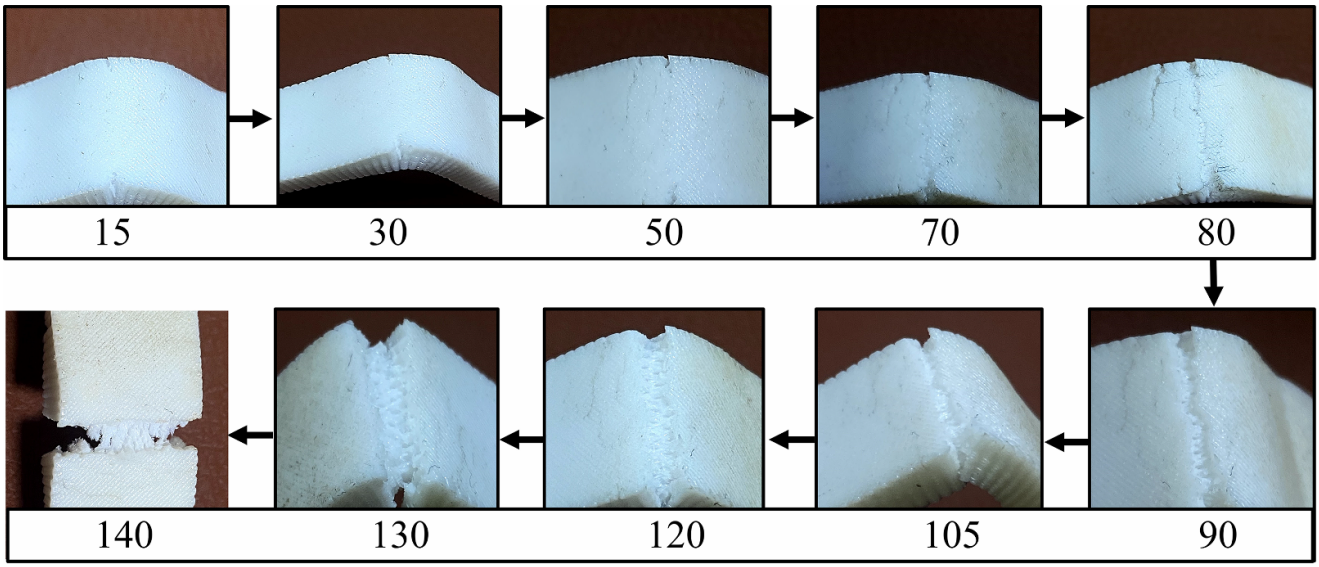


Figure 11. Photomacrograph of fractured surface at various actuation cycles for PDP sample.

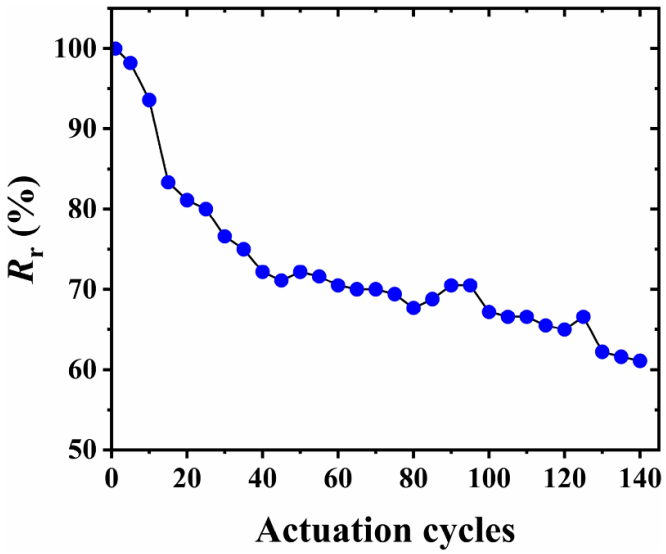


Figure 12. R_r vs. actuation cycles plot for shape recovery cycle life measurements of PDP sample.

increment of the R_r, but subsequently, it decreases due to the functional fatigue being prevalent.

4.6. Fractography study by field emission-SEM

A study of the fractured surface's morphology helps understand the fracture mechanism. From figure 13, it can be clearly observed that the morphology of the printed part is a layered structure common for additively manufactured samples. Further magnification of a single fiber top removed by delamination shows the polymer chips created due to continuous bending and unbending during the shape recovery cycles. In figure 14, the fractured surface FE-SEM micrographs reveal stretched fibers in the layers suggesting a ductile failure. Further magnification of the fiber cross-section reveals nanopores from which the crack initiation may have occurred. These pores may have been developed due to the vapors entrapped during continuous heating and rapid cooling during the printing.

4.7. Working prototypes showing the difference between PDP and PAP methodologies

The results of the detailed study and characterization between different programming methodologies have been utilized to formulate separate shape recovery schemes for PDP and PAP, as shown in figure 15. Then the working prototypes were developed and tested in hot water actuation apparatus, as shown in figure 16. The PAP prototype curls within around 11 s as per the scheme shown in figure 15(b). The PDP sample uncurls the temporary deformation within 13 s as per the scheme depicted in figure 15(a). The working prototypes for

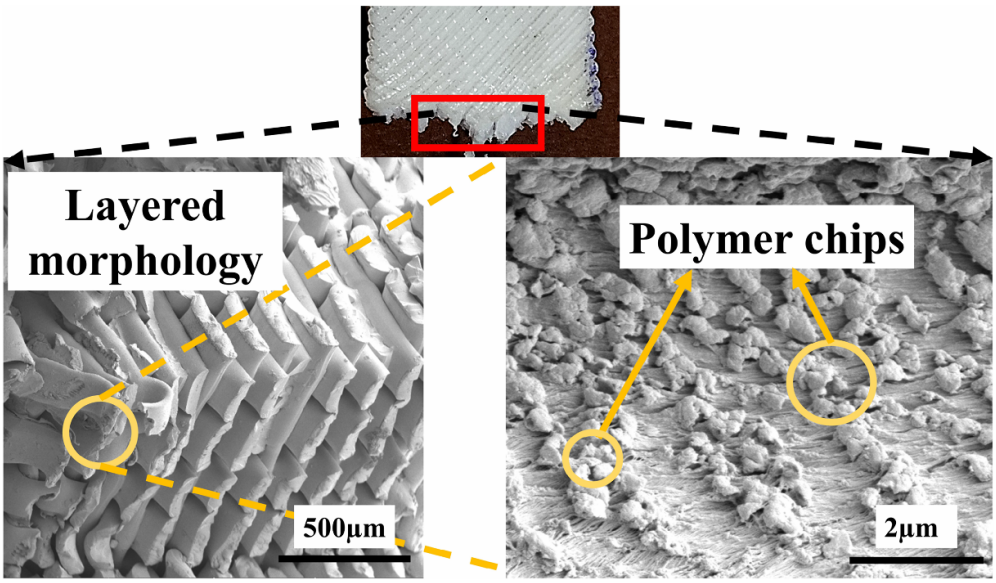


Figure 13. FE-SEM micrograph of the fracture surface showing the layered morphology and the polymer chips generated due to actuation cycles.

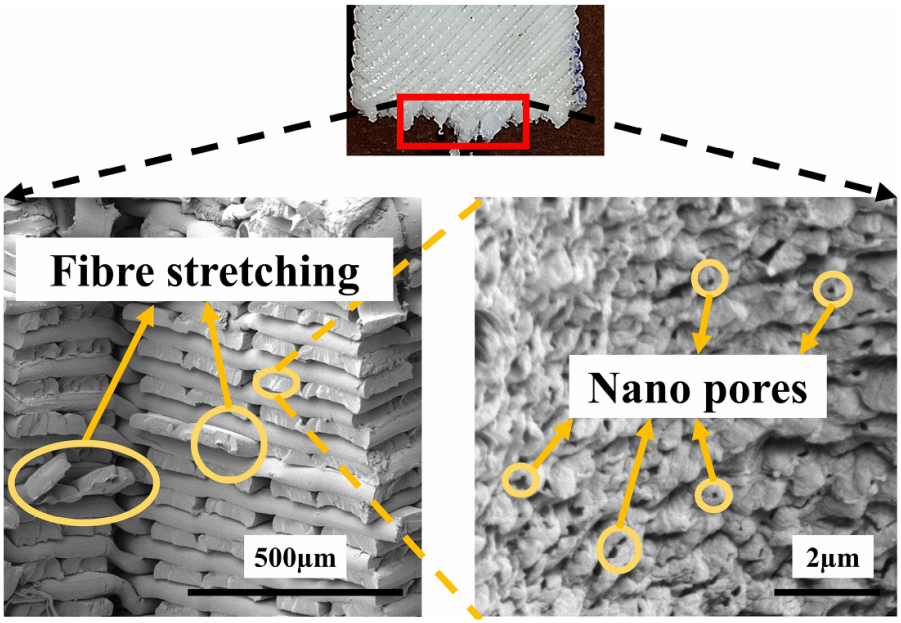


Figure 14. FE-SEM micrograph of the fracture surface showing the fiber stretching and the nanopores.

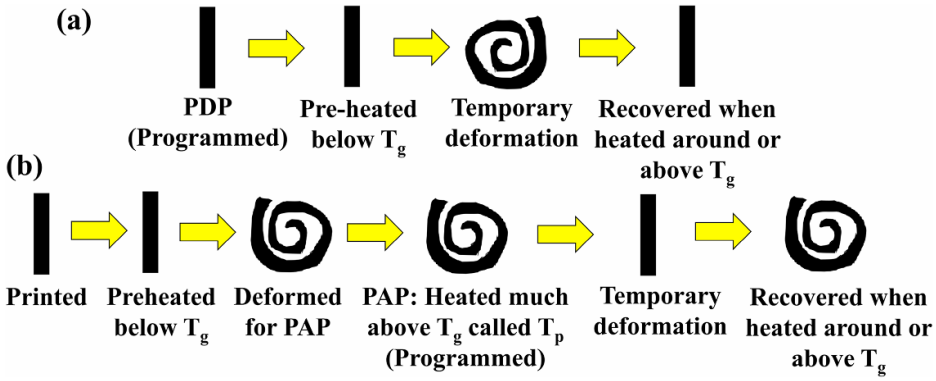


Figure 15. Material and shape programming schemes (a) PDP (b) PAP.

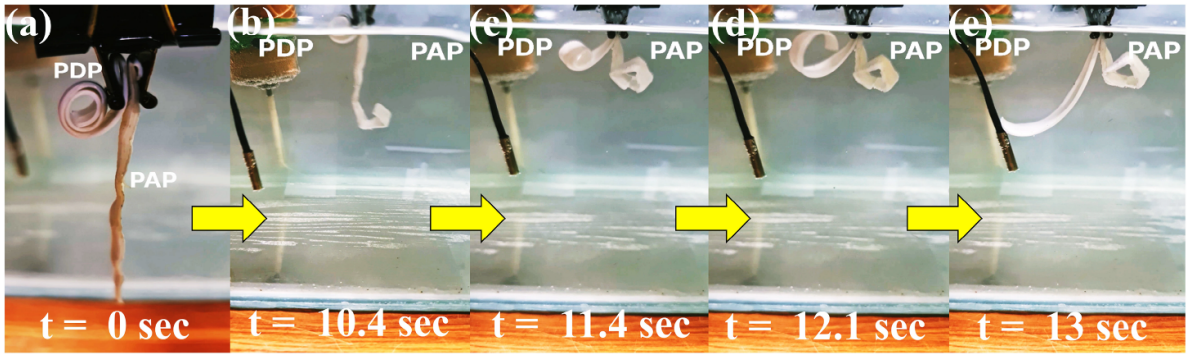


Figure 16. Working prototype showing the effectiveness of PDP and PAP for programming the same SMP in two different ways as per the scheme developed.

both types of programming are actuated side by side for a better comparison of the shape recovery, as in figure 16.

5. Conclusion

In this study, we investigated the effects of two novel programming methodologies, PDP and PAP, on the shape memory and other material properties of an FDM 4D printed SMP. The results were supported by multiple characterization techniques, including x-ray diffraction, nanoindentation, ATR-FTIR spectroscopy, and FE-SEM. While both programming methods had individual advantages and disadvantages, the results demonstrated that PDP outperformed PAP in terms of recovery ratio, fixation ratio, and SMI. PDP exhibited superior shape memory characteristics owing to its higher level of pre-strain stored during the printing process than PAP, which had a relatively lower pre-strain during programming in the post-processing stage. The x-ray diffraction results revealed that PDP and PAP exhibited distinguishing peak characteristics, indicating a disparity in crystallinity percentage (6.5% for PDP and 42% for PAP). In the ATR-FTIR analysis, it is found that the amorphous peak shifted from 855 cm^{-1} for PDP to 871 cm^{-1} for PAP, indicating the formation of a crystalline structure due to the dipole–dipole interaction in the crystal lattice in PAP, thus showing a higher crystallinity index (78.3%) than PDP (59%). Micromechanical characterization like nanoindentation analysis reveals that PDP samples exhibited lower hardness and Young's modulus values than PAP samples. The recovery cycle life test showed that the PDP sample could actuate up to 140 cycles before final failure, providing recovery of around 60%, making it viable for low-cost, low-cycle thermally actuated shape-changing structures for various 4D printing applications. As an example, a working prototype was developed to demonstrate PAP and PDP side-by-side actuating in hot water based on a pre-determined scheme. By combining tailored shape memory properties through the proposed programming methodologies along with the affordable and widely accessible FDM 4D printing technique, we expect that it will have a widespread application in the field of biomedical, photonics, and robotics to create economically feasible self-transforming and adaptable structures.

We can fairly conclude that the present work contributes to the advancement of smart materials and structures through 4D printing by guiding researchers and industry professionals in selecting the most appropriate programming strategy for their specific needs.

Data availability statement

The data cannot be made publicly available upon publication because they contain commercially sensitive information. The data that support the findings of this study are available upon reasonable request from the authors.

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Conflict of interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

[1] Kruth J-P, Leu M C and Nakagawa T 1998 Progress in additive manufacturing and rapid prototyping *CIRP Ann.* **47** 525–40

[2] Manogharan G, Wysk R A and Harrysson O L A 2016 *Additive Manufacturing–Integrated Hybrid Manufacturing and Subtractive Processes: Economic Model and Analysis* vol 29 (Taylor &: Francis)

[3] Gibson I, Rosen D and Stucker B 2015 Development of additive manufacturing technology *Additive Manufacturing Technologies: 3D Printing, Rapid Prototyping, and Direct Digital Manufacturing* (New York: Springer) pp 19–42

[4] Attaran M 2017 The rise of 3D printing: the advantages of additive manufacturing over traditional manufacturing *Bus. Horiz.* **60** 677–88

[5] Samal B B, Jena A, Varshney S K and Kumar C S 2023 4D printing: an experimental case study on processing of shape memory polymer by FDM/FFF for nature inspired structures *Advances in Additive Manufacturing Artificial Intelligence, Nature-Inspired, and Biomanufacturing* ed A Kumar, R K Mittal and A Haleem (Amsterdam: Elsevier) pp 361–77

[6] Choi J, Kwon O-C, Jo W, Lee H J and Moon M-W 2015 4D printing technology: a review *3D Print. Addit. Manuf.* **2** 159–67

[7] Samal B B, Jena A, Varshney S K and Kumar C S 2023 FDM 4D printing: a low-cost approach of shape programming and assessing the shape memory properties using angle measurement methods in hot water actuation testing apparatus *Mater. Today Proc.* (https://doi.org/10.1016/j.matpr.2023.02.038)

[8] Kuang X, Yue L and Qi H J 2023 Introduction to 4D printing: concepts and material systems *Additive Manufacturing Technology* (Berlin: Wiley) pp 1–42

[9] Pei E 2014 4D printing: dawn of an emerging technology cycle *Assem. Autom.* **34** 310–4

[10] Gladman A S, Matsumoto E A, Nuzzo R G, Mahadevan L and Lewis J A 2016 Biomimetic 4D printing *Nat. Mater.* **15** 413

[11] Tibbits S 2014 4D printing: multi-material shape change *Archit. Des.* **84** 116–21

[12] Mao Y, Yu K, Isakov M S, Wu J, Dunn M L and Jerry Qi H 2015 Sequential self-folding structures by 3D printed digital shape memory polymers *Sci. Rep.* **5** 1–8

[13] Wu J, Yuan C, Ding Z, Isakov M, Mao Y, Wang T, Dunn M L and Qi H J 2016 Multi-shape active composites by 3D printing of digital shape memory polymers *Sci. Rep.* **6** 24224

[14] Samal B B, Jena A and Mishra D 2017 Controlled shape changing components by using 4D printing technology *10th International Conference on Precision, Meso, Micro and Nano Engineering (COPEN 10) IIT, Madras* 2017 78–81 (available at: www.copen.ac.in/proceedings/copen10/copen/19.revised_manuscript_262.pdf)

[15] Samal B B, Jena A and Mishra D 2018 Indian researchers develop world's strongest 4D printed parts *Manufactur3D Magazine* (available at: https://manufactur3dmag.com/indian-researchers-develop-worlds-strongest-4d-printed-parts/) (Accessed 28 January 2020)

[16] Meisel N A, Elliott A M and Williams C B 2015 A procedure for creating actuated joints via embedding shape memory alloys in PolyJet 3D printing *J. Intell. Mater. Syst. Struct.* **26** 1498–512

[17] Akbari S, Sakhaei A H, Panjwani S, Kowsari K and Ge Q 2021 Shape memory alloy based 3D printed composite actuators with variable stiffness and large reversible deformation *Sens. Actuators* A **321** 112598

[18] Lalegani Dezaki M and Bodaghi M 2023 Magnetorheological elastomer-based 4D printed electroactive composite actuators *Sens. Actuators* A **349** 114063

[19] Behl M and Lendlein A 2007 Shape-memory polymers *Mater. Today* **10** 20–28

[20] Hosseinzadeh M, Ghoreishi M and Narooei K 2023 4D printing of shape memory polylactic acid beams: an experimental investigation into FDM additive manufacturing process parameters, mathematical modeling, and optimization *J. Manuf. Process.* **85** 774–82

[21] Aberoumand M *et al* 2023 4D printing of polyvinyl chloride (PVC): a detailed analysis of microstructure, programming, and shape memory performance *Macromol. Mater. Eng.* **308** 2200677

[22] Kuang X, Chen K, Dunn C K, Wu J, Li V C F and Qi H J 2018 3D printing of highly stretchable, shape-memory, and self-healing elastomer toward novel 4D printing *ACS Appl. Mater. Interfaces* **10** 7381–8

[23] Ding Z, Yuan C, Peng X, Wang T, Qi H J and Dunn M L 2017 Direct 4D printing via active composite materials *Sci. Adv.* **3** e1602890

[24] Raviv D *et al* 2014 Active printed materials for complex self-evolving deformations *Sci. Rep.* **4** 7422

[25] Ding Z, Weeger O, Qi H J and Dunn M L 2018 4D rods: 3D structures via programmable 1D composite rods *Materials & Design* **137** 256–65

[26] Ge Q, Sakhaei A H, Lee H, Dunn C K, Fang N X and Dunn M L 2016 Multimaterial 4D printing with tailorable shape memory polymers *Sci. Rep.* **6** 31110

[27] Kokkinis D, Schaffner M and Studart A R 2015 Multimaterial magnetically assisted 3D printing of composite materials *Nat. Commun.* **6** 8643

[28] Monzón M D, Paz R, Pei E, Ortega F, Suárez L A, Ortega Z, Alemán M E, Plucinski T and Clow N 2017 4D printing: processability and measurement of recovery force in shape memory polymers *Int. J. Adv. Manuf. Technol.* **89** 1827–36

[29] Ohki T, Ni -Q-Q, Ohsako N and Iwamoto M 2004 Mechanical and shape memory behavior of composites with shape memory polymer *Composites* A **35** 1065–73

[30] Bodaghi M, Damanpack A R and Liao W H 2016 Self-expanding/shrinking structures by 4D printing *Smart Mater. Struct.* **25** 105034

[31] Bodaghi M, Damanpack A R and Liao W H 2017 Adaptive metamaterials by functionally graded 4D printing *Mater. Des.* **135** 26–36

[32] Ly S T and Kim J Y 2017 4D printing—fused deposition modeling printing with thermal-responsive shape memory polymers *Int. J. Precis. Eng. Manuf.—Green Technol.* **4** 267–72

[33] Ortega F, Versino F, López O V and García M A 2022 Biobased composites from agro-industrial wastes and by-products *Emerg. Mater.* **5** 873–921

[34] Farah S, Anderson D G and Langer R 2016 Physical and mechanical properties of PLA, and their functions in widespread applications—a comprehensive review *Adv. Drug Deliv. Rev.* **107** 367–92

[35] Leist S K, Gao D, Chiou R and Zhou J 2017 Investigating the shape memory properties of 4D printed polylactic acid (PLA) and the concept of 4D printing onto nylon fabrics for the creation of smart textiles *Virtual Phys. Prototyp.* **12** 290–300

[36] Zhang F, Wang L, Zheng Z, Liu Y and Leng J 2019 Magnetic programming of 4D printed shape memory composite structures *Composites* A **125** 105571

[37] Ma S, Jiang Z, Wang M, Zhang L, Liang Y, Zhang Z, Ren L and Ren L 2021 4D printing of PLA/PCL shape memory composites with controllable sequential deformation *Bio-Des. Manuf.* **4** 867–78

[38] Barletta M, Gisario A and Mehrpouya M 2021 4D printing of shape memory polylactic acid (PLA) components: investigating the role of the operational parameters in fused deposition modelling (FDM) *J. Manuf. Process.* **61** 473–80
[39] Shin S and So H 2020 Effect of 3D printing raster angle on reversible thermo-responsive composites using PLA/paper bilayer *Smart Mater. Struct.* **29** 105016
[40] Senatov F S, Niaza K V, Zadorozhnyy M Y, Maksimkin A V, Kaloshkin S D and Estrin Y Z 2016 Mechanical properties and shape memory effect of 3D-printed PLA-based porous scaffolds *J. Mech. Behav. Biomed. Mater.* **57** 139–48
[41] Cheng T, Tahouni Y, Wood D, Stolz B, Mülhaupt R and Menges A 2020 Multifunctional Mesostructures: Design and Material Programming for 4D-printing *Proc. of the 5th Annual ACM Symp. on Computational Fabrication* SCF '20 (New York: Association for Computing Machinery) 1–10
[42] Hu G F, Damanpack A R, Bodaghi M and Liao W H 2017 Increasing dimension of structures by 4D printing shape memory polymers via fused deposition modeling *Smart Mater. Struct.* **26** 125023
[43] Bodaghi M, Damanpack A R and Liao W H 2018 Triple shape memory polymers by 4D printing *Smart Mater. Struct.* **27** 065010
[44] Lendlein A and Kelch S 2002 Shape-memory polymers *Angew. Chem., Int. Ed.* **41** 2034–57
[45] Li G and Wang A 2016 Cold, warm, and hot programming of shape memory polymers *J. Polym. Sci.* B **54** 1319–39
[46] Chandalia R and Jain P WOL 3D Pvt. Ltd (available at: https://worldoflilliputs.com/) (Accessed 10 September 2022)
[47] Carlson M and Li Y 2020 Development and kinetic evaluation of a low-cost temperature-sensitive shape memory polymer for 4-dimensional printing *Int. J. Adv. Manuf. Technol.* **106** 4263–79
[48] Wang Y and Li X 2020 An accurate finite element approach for programming 4D-printed self-morphing structures produced by fused deposition modeling *Mech. Mater.* **151** 103628
[49] Brown D, Christian W and Hanson R M Tracker video analysis and modeling tool for physics education (available at: https://physlets.org/tracker/) (Accessed 10 September 2022)
[50] Siegel A F 2012 Hypothesis testing: deciding between reality and coincidence *Practical Business Statistics* 6th edn, ed A F Siegel (Boston: Academic) ch 10, pp 249–87
[51] Yang X-S 2017 Regression and curve fitting *Engineering Mathematics with Examples and Applications* ed X-S Yang (New York: Academic) ch 19, pp 215–28
[52] Carroll R J and Schneider H 1985 A note on Levene's tests for equality of variances *Stat. Probab. Lett.* **3** 191–4
[53] Oliver W C and Pharr G M 1992 An improved technique for determining hardness and elastic modulus using load and displacement sensing indentation experiments *J. Mater. Res.* **7** 1564–83
[54] Choong Y Y C, Maleksaeedi S, Eng H, Wei J and Su P C 2017 4D printing of high performance shape memory polymer using stereolithography *Mater. Des.* **126** 219–25
[55] Liao Y, Liu C, Coppola B, Barra G, Di Maio L, Incarnato L and Lafdi K 2019 Effect of porosity and crystallinity on 3D printed PLA properties *Polymers* **11** 1487
[56] Echeverría C, Limón I, Muñoz-Bonilla A, Fernández-García M and López D 2021 Development of highly crystalline polylactic acid with β-crystalline phase from the induced alignment of electrospun fibers *Polymers* **13** 2860
[57] Riba J R, Cailloux J, Cantero R, Canals T and Maspoch M L 2018 Multivariable methods applied to FTIR: a powerful technique to highlight architectural changes in poly(lactic acid) *Polym. Test.* **65** 264–9
[58] Xu R, Xie J and Lei C 2017 Influence of melt-draw ratio on the crystalline behaviour of a polylactic acid cast film with a chi structure *RSC Adv.* **7** 39914–21
[59] Na B, Tian N, Lv R, Li Z, Xu W and Fu Q 2010 Evidence of sequential ordering during cold crystallization of poly (l-lactide) *Polymer* **51** 563–7